TractoInferno Validation Report

csttool Corticospinal Tract Extraction — Validation Against TractoInferno PYT Reference Bundles

1. Validation Design

1.1 Objective

Evaluate the spatial agreement between corticospinal tract (CST) bundles extracted by **csttool** and reference pyramidal tract (PYT) bundles from the **TractoInferno** dataset (OpenNeuro ds003900). The validation quantifies how well an automated, deterministic, atlas-based extraction pipeline recovers a known tract across a large, heterogeneous cohort.

1.2 Reference Dataset

TractoInferno provides expert-segmented white matter bundles for 284 subjects. The **trainset** partition was used, containing reference PYT bundles generated via probabilistic tractography with expert-guided ROI selection. Each subject includes left and right PYT bundles in `.trk` format and a fractional anisotropy (FA) image defining the anatomical reference space.

1.3 Candidate Generation

For each subject, csttool's full pipeline was executed with the following parameters:

Parameter	Value	Description
Denoising	NLMeans	Non-local means denoising
Preprocessing	Enabled	Denoising + brain masking
Direction model	CSA-ODF (SH order 6)	Constant solid angle ODF
Tracking	Deterministic	Single-peak (npeaks=1)
Step size	0.5 mm	Streamline propagation step
FA threshold	0.2	Tracking termination
Seed density	1 seed/voxel	In white matter mask
Peak threshold	0.8 (relative)	Relative peak threshold
Min separation angle	45°	Between ODF peaks
ROI approach	Atlas-to-subject	Harvard-Oxford via ANTs SyN
Field strength	3.0 T	Assumed acquisition field strength

1.4 Validation Contract

A formal validation contract governs the comparison process. Each subject–hemisphere pair is assigned exactly one outcome status:

Status	Condition	Metrics
OK	Spatial checks pass, candidate non-empty, no structural warnings	Computed
DEGRADED	Spatial checks pass, but candidate empty or below minimum streamline count	Computed (may be zero)

INVALID_SPACE	Spatial compatibility violated (affine mismatch, grid incompatibility, or candidate missing)	Not computed

INVALID_SPACE cases are excluded from quantitative analysis and reported separately. DEGRADED cases remain in the analysis but are flagged. This separation ensures that infrastructure failures do not contaminate algorithmic performance assessment.

1.5 Spatial Compatibility Checks

Before computing any metric, the validator verifies that candidate and reference bundles share a compatible coordinate system:

- The candidate tractogram must be mappable into the reference FA space.
- Affine matrices must agree within a tolerance of 0.5 mm (RMS).
- Voxel grid dimensions and orientation must match the reference anatomy.

If any check fails, the comparison is recorded as INVALID_SPACE with an explicit error code (e.g., `E_CAND_MISSING`).

2. Metrics

2.1 Streamline Voxelization

Both candidate and reference streamlines are voxelized onto the reference FA image grid (1×1×1 mm^3 isotropic) using DIPY's `density_map()` function. The resulting density maps are binarized: any voxel traversed by at least one streamline is set to 1, all others to 0. This produces two binary masks, *A* (candidate) and *B* (reference).

2.2 Dice Coefficient

The Dice coefficient measures the spatial overlap between two binary volumes. It is the harmonic mean of precision and recall:

```
Dice = 2 × |A ∩ B| / (|A| + |B|)
```

where |A ∩ B| is the number of voxels present in both masks, and |A|, |B| are the total occupied voxels in each mask. Dice ranges from 0 (no overlap) to 1 (perfect overlap). It is sensitive to both the position and size of the masks.

2.3 Overlap Metrics (Precision and Recall)

The Dice coefficient can be decomposed into directional overlap measures:

Metric	Formula	Interpretation
Precision (overlap_cand_in_ref)	`|A ∩ B| / |A|`	Fraction of candidate voxels that fall within the reference. High precision means the extraction is spatially accurate—it places streamlines where they belong.
Recall (overlap_ref_in_cand)	`|A ∩ B| / |B|`	Fraction of reference voxels captured by the candidate. High recall means the extraction covers the full extent of the reference bundle.

The relationship between these metrics and Dice is:

```
Dice = 2 / (1/Precision + 1/Recall)
```

A method with high precision but low recall produces conservative, spatially accurate bundles that capture the core of the tract but miss peripheral streamlines. This is the expected behavior of deterministic tractography compared to probabilistic references.

2.4 Streamline Count Metrics

Metric	Formula	Interpretation
Candidate count	Number of streamlines in csttool output	Size of the extracted bundle
Reference count	Number of streamlines in TractoInferno PYT	Size of the reference bundle
Count ratio	Candidate / Reference	Relative bundle density

2.5 Supplementary Voxel Counts

Each validation result records the number of reference voxels, candidate voxels, and intersection voxels, as well as the number of mapped streamline points and any out-of-bounds points. These support debugging and deeper analysis.

3. Results

3.1 Cohort Summary

167	334	98.8%
SUBJECTS	HEMISPHERE COMPARISONS	SUCCESS RATE

4

INVALID (EXCLUDED)

Of 334 hemisphere comparisons (167 subjects × 2), 330 yielded valid metrics (status OK) and 4 were excluded due to spatial incompatibility (INVALID_SPACE, error code E_CAND_MISSING). No DEGRADED cases were observed.

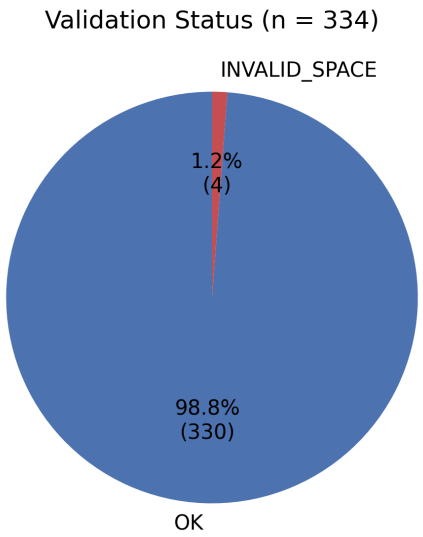

Figure 1: Validation outcome distribution.

3.2 Dice Coefficient

Statistic	Value
Mean ± SD	0.283 ± 0.065
Median	0.285
Range	[0.053, 0.471]
Left hemisphere (mean ± SD)	0.268 ± 0.066
Right hemisphere (mean ± SD)	0.299 ± 0.060

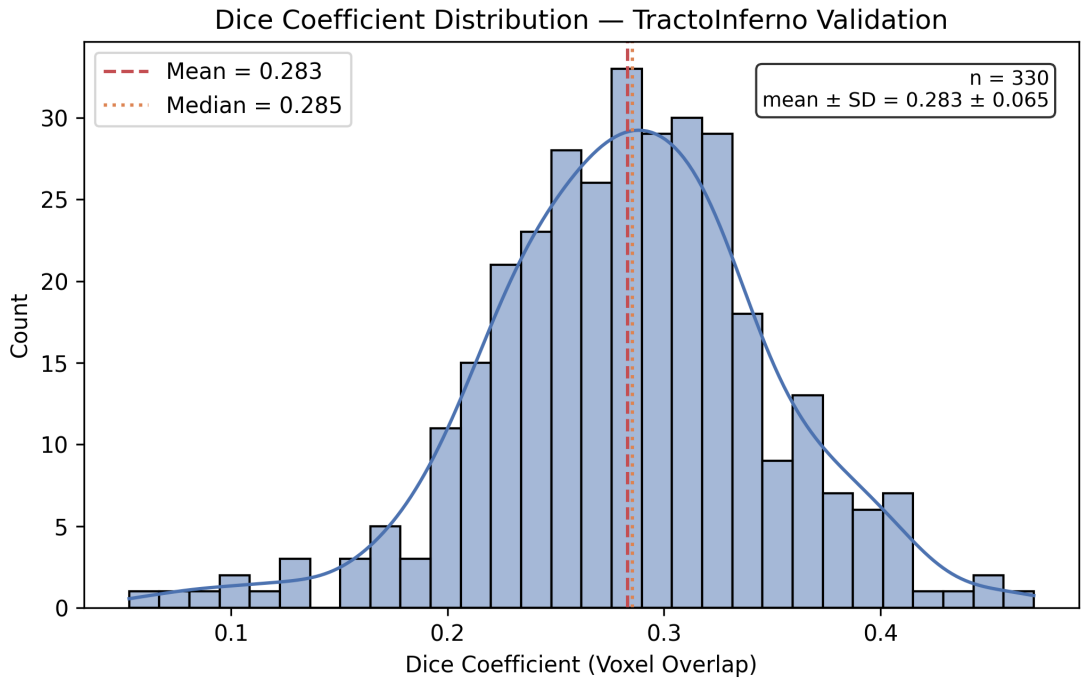

Figure 2: Distribution of Dice coefficients across 330 valid hemisphere comparisons.

3.3 Precision and Recall

Metric	Mean ± SD	Interpretation
Precision (o_cand_in_ref)	0.854 ± 0.066	85.4% of extracted voxels are within the reference
Recall (o_ref_in_cand)	0.172 ± 0.047	17.2% of reference voxels are captured

The high precision and low recall pattern is characteristic of deterministic tractography validated against probabilistic references. csttool's deterministic tracking with a single ODF peak produces spatially focused, conservative bundles that accurately capture the core of the CST but do not extend to the full probabilistic envelope of the reference. This is a methodological property, not a deficiency—the tool prioritizes specificity over sensitivity, which is appropriate for clinical applications where false-positive tract assignment carries greater risk than under-coverage.

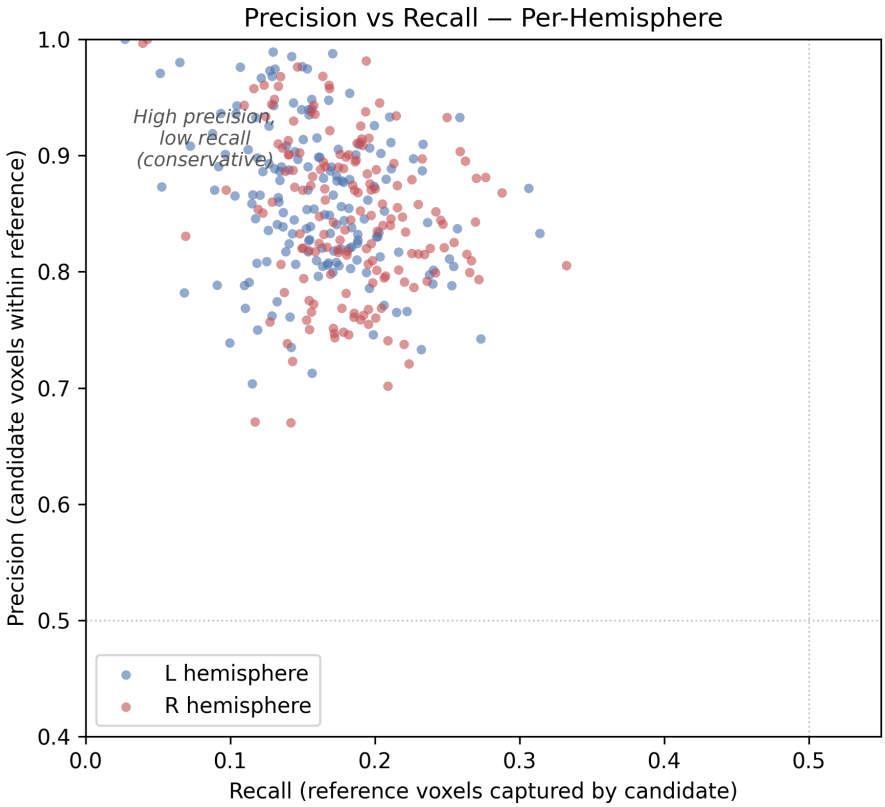

Figure 3: Precision vs. recall for each hemisphere comparison. The clustering in the upper-left quadrant confirms conservative, high-precision extraction.

3.4 Hemisphere Comparison

The right hemisphere shows slightly higher Dice coefficients than the left (0.299 vs. 0.268), which may reflect anatomical asymmetries in the pyramidal tract or differences in how the reference bundles were segmented.

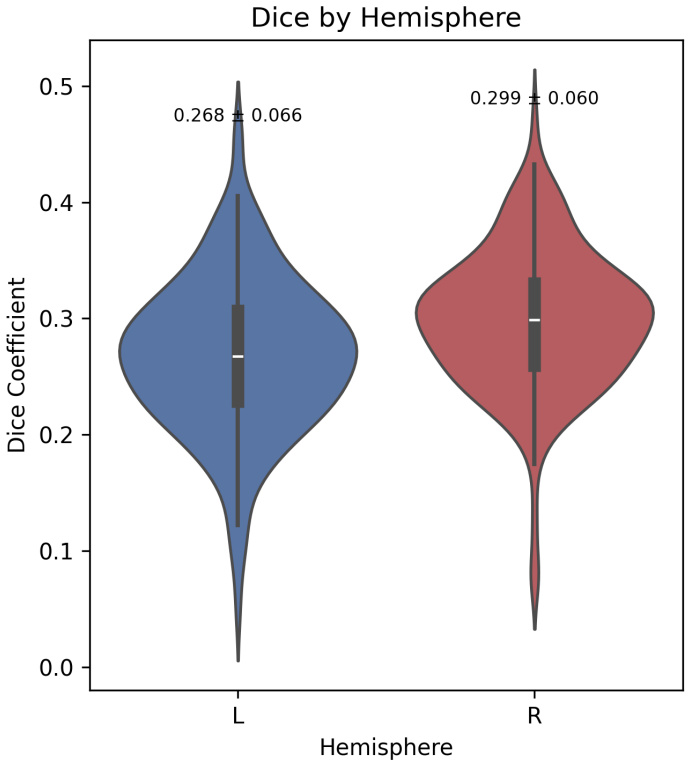

Figure 4: Dice coefficient distribution by hemisphere.

3.5 Streamline Counts

	Mean	Range
Candidate (csttool)	4,093	358 – 9,676
Reference (TractoInferno)	61,188	19,244 – 169,699

csttool extracts approximately 7× fewer streamlines than the probabilistic reference bundles. This ratio reflects the fundamental difference between deterministic single-seed tracking (1 seed/voxel, single peak) and probabilistic multi-seed approaches. The reduced streamline count is consistent with the low recall and high precision pattern: fewer streamlines concentrated in the tract core.

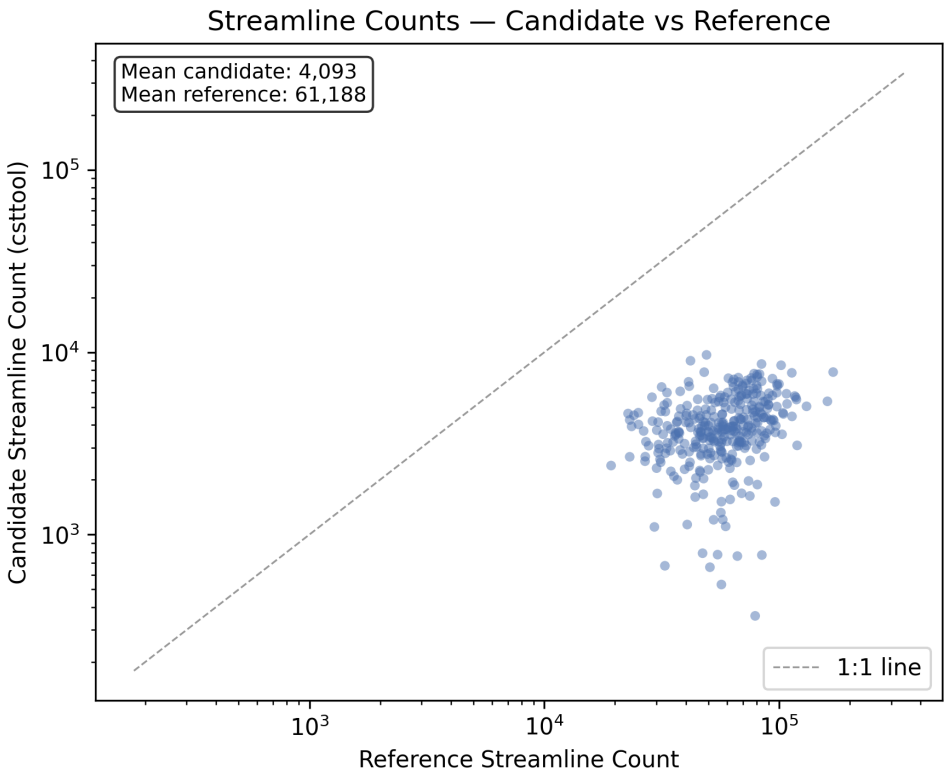

Figure 5: Candidate vs. reference streamline counts on a log scale. The dashed line indicates 1:1 correspondence.

3.6 Subject-Level Paired Comparison

The preceding analyses aggregate across hemispheres, which can mask within-subject variability. To assess whether performance is consistent within individual subjects, Dice, precision, and recall were plotted as paired L/R values for each subject.

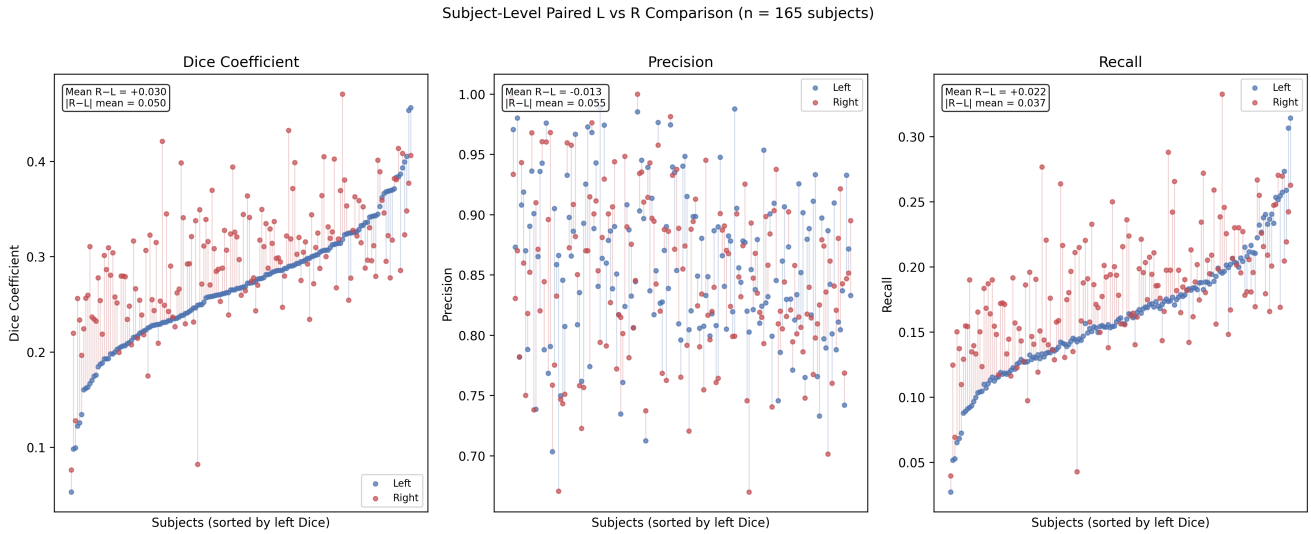

Figure 6: Subject-level paired L vs. R comparison for Dice coefficient, precision, and recall. Each subject contributes two points (blue = left, red = right) connected by a vertical line. Subjects are sorted by left-hemisphere Dice. Short connecting lines indicate consistent within-subject performance.

Key observations:

- **Within-subject consistency:** Left and right hemispheres track each other closely across subjects. The mean absolute L–R difference is 0.050 for Dice, 0.055 for precision, and 0.037 for recall, indicating that performance is stable within subjects rather than driven by one hemisphere.
- **Slight right-hemisphere advantage:** The mean R–L difference is +0.030 for Dice and +0.022 for recall, consistent with the violin plot in Section 3.4. This trend is uniform across subjects rather than driven by outliers.
- **Outlier identification:** A small number of subjects show low Dice in both hemispheres simultaneously (lower-left of the Dice panel), suggesting subject-level data quality issues rather than hemisphere-specific failures.
- **Robustness:** The monotonic rise in the sorted Dice panel demonstrates a continuous performance distribution with no bimodal clustering, supporting the claim that the pipeline generalizes across the cohort.

4. Methodology Notes

4.1 Why Dice Appears Low

A Dice coefficient of ~0.28 may appear low relative to segmentation tasks where values above 0.7 are expected. However, tractography bundle comparison differs from volumetric segmentation in several important ways:

- **Methodological asymmetry:** The candidate (deterministic) and reference (probabilistic) bundles are generated with fundamentally different algorithms, producing different spatial extents by design.
- **Volume asymmetry:** Reference bundles occupy ~5–15× more voxels than candidates, which mathematically depresses Dice even when the candidate is fully contained within the reference.
- **Dice sensitivity to size mismatch:** When $|A| << |B|$, even perfect precision (all of A inside B) yields Dice $\approx 2|A|/(|A|+|B|)$, which is dominated by the size ratio.

The precision–recall decomposition provides a more informative characterization than Dice alone: 85.4% precision with 17.2% recall demonstrates that the extraction is spatially accurate but deliberately conservative.

4.2 Reproducibility

All results were generated from a single batch run (run ID: `2026-02-01_git-1433ac4`, git commit `1433ac4`) with fixed parameters and no manual intervention. The deterministic tracking algorithm and fixed random seed ensure identical results across reruns on the same data. Each subject's validation result is stored as a structured JSON file conforming to a versioned schema (`csttool_validation_result_schema_v1.json`).

4.3 Limitations

- Validation is limited to TractoInferno's trainset; no clinical cohort with known pathology was available for validation.
- The reference PYT bundles are themselves algorithmic outputs (probabilistic tractography with expert ROIs), not ground truth anatomical dissections.
- Two subjects (sub-1025, sub-1074) failed spatial compatibility checks, likely due to data quality issues in the source dataset.